

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – «from SIH portal»

Problem Statement Title – AR-Based Vocational Training

Simulator for Industrial Safety in Jharkhand's Mining &

Manufacturing Sector

Theme – Smart Education

PS Category – Software

Team ID – «from SIH portal»

Team Name – Technova



SURAKSHAAR — OFFLINE AR SAFETY TRAINING



What it is

- An offline AR safety-training and certification platform running on any Android 10+ phone — no headset, no ARCore, no internet at any point.
- The worker sweeps the phone across their workplace; the app finds the floor, the clear space and real doorways, then builds a mine gallery fitted to that room.
- Five DGMS-aligned domains: Fire & Explosion, Gas Leak & Confined Space, Machinery & Lockout-Tagout, Roof Fall & Working at Height, Electrical & First Aid.
- Drills score what the worker does under time pressure — action sequencing, reaction time, PPE selection — not attendance.

Why it is different

- Runs on the low-cost handset a contract worker already carries. ARCore-based AR needs certified hardware most budget phones do not have.
- The certificate is a cryptographically signed QR code any inspector can verify offline, on a fresh install, with no account and no network.
- Audio-first throughout, so a worker who reads slowly or not at all can still run the drill. English ships today; the string layer is script-agnostic and Hindi and Santali (Ol Chiki) are the next translation pass.
- Genuinely offline end to end — exercised in airplane mode; no worker data ever leaves the handset.
- A working release APK backed by 189 automated tests, not a mock-up.

Platform and stack

- Flutter 3.44 / Dart — one Android APK, minSdk 29 (Android 10 and above).
- Kotlin platform channel reading
TYPE_GAME_ROTATION_VECTOR: gyroscope plus accelerometer, magnetometer deliberately excluded because magnetic heading is unusable underground and beside steel-plant motors.
- Madgwick AHRS filter in Dart as the fallback for handsets with no fused sensor.
- Camera2 lens metadata gives the true field of view, so overlays stay aligned instead of sliding as the worker turns.
- SQLite on the phone, with signed record bundles as the export format. The supervisor dashboard that consumes them is specified and not yet built; the bundle is what the phone already produces.

How the AR works

- Gravity-referenced pose makes the floor plane known rather than estimated — only the camera height has to be measured.
- Room sensing: each camera frame is reduced to a 32×64 grid and traced upward from the worker's feet, using brightness and texture together so a grey wall above a grey floor is still detected. Tall rather than square on purpose: rows are what set how precisely a doorway can be measured.
- Door detection without machine learning: vertical edge pairs meeting the floor are accepted only if they measure about 0.9 m by 2 m in real metres. Geometry does the semantic filtering, so no model weights ship in the APK.
- All content is procedural vector art — no 3D assets, no licensing risk, and a steady frame rate on mid-range hardware.
- Certificate: CBOR, zlib, Ed25519 signature and base45 QR, following the approach India already uses for DIVOC and CoWIN credentials.

Why it is feasible

- The hardware requirement is a phone the workforce already owns — no headset purchase, no ARCore certification, no data plan.
- A release APK builds today, with 189 automated tests covering projection mathematics, environment detection, adaptive assessment and certificate cryptography.
- Nothing in the design needs a cloud contract or a per-site recurring cost: records move as signed files, and the planned supervisor dashboard is a laptop on a local hotspot.
- Fits existing statutory practice under the Mines Act 1952 and the Factories Act 1948 rather than asking regulators to adopt something new.

Risks, and how they are handled

- Magnetic interference underground — handled by magnetometer-free sensor fusion, chosen for exactly this reason.
- Rotation-only tracking rather than SLAM — drills are designed around looking around, and the simulated gallery is fitted to the measured room so nothing is placed inside a real wall.
- Santali is a low-resource language — every translated string is badged for native review with a Hindi fallback before it is shown, so no unverified safety wording is ever presented as authoritative.
- Device fragmentation — an in-app diagnostics screen reads Android's own process-exit records, so a fault at a remote site is diagnosable without a laptop.
- Falsified training records — every entry is signed by the device key and counter-signed by the organisation before a certificate counts as verified.

Impact

- Jharkhand's coal, steel and mica workforce — hundreds of thousands of workers, many of them young tribal recruits with no prior industrial exposure.
- DGMS Dhanbad recorded 48 fatal mine accidents in Jharkhand in 2022–23, a large share involving workers with under 30 days of orientation.
- Classroom training on static manuals retains under 20% after one week; practising the drill with visible consequences targets exactly that gap.
- Signed records carry each worker's days-since-induction, so the under-30-day cohort can be surfaced by name and acted on before an accident rather than after.

Benefits

- Social — audio-first delivery, with translation into the worker's own language, reaching the contract labour that headset programmes exclude entirely.
- Economic — effectively no marginal cost per worker trained; avoided accidents reduce compensation, downtime and lost production.
- Regulatory — certification that evidences comprehension rather than attendance, and that an inspector can verify offline at the pit head.
- Environmental — no headset hardware to manufacture, ship, power or dispose of.
- Scale — works for the small mines and contractors that could never fund a simulator room, which is where the orientation gap is widest.

Statutory and domain sources

- Directorate General of Mines Safety, Dhanbad — annual statistics of mine accidents and standard notes on causation.
- The Mines Act 1952, the Mines Rules 1955 and the Mines Vocational Training Rules 1966 — periodic training and certification obligations.
- The Factories Act 1948 — safety training duties for steel and manufacturing units.
- DGMS circulars on confined-space entry, gas testing, and support of roof and sides.
- ISO 7010 — registered safety signs, followed for every pictogram in the app.

Technical references

- S. Madgwick, An efficient orientation filter for inertial and inertial/magnetic sensor arrays, 2010 — basis of the fallback pose filter.
- DIVOC and CoWIN verifiable credential design, Government of India — model for the offline-verifiable QR certificate.
- RFC 8032 (Ed25519 signatures), RFC 8949 (CBOR) and Base45 encoding for compact QR payloads.
- Google ML Kit — on-device barcode scanning, used with no network dependency.
- Project source and build: github.com/sriram4n/surakshaar